Problem 19

Determine the random event X if:

$$1^\circ\ A + X = A + B; \quad 2^\circ\ AB + X = (A + C)(B + C)$$

Solution

Part 1°: Solve $A + X = A + B$ for X

Method 1: Direct Manipulation

Starting with:

$$A + X = A + B$$

We want to isolate X. Intuitively, X should be the "part of B not already in A".

Step 1: Intersect Both Sides with $\bar{A}$

$$\bar{A}(A + X) = \bar{A}(A + B)$$

Using distributive law:

$$\bar{A}A + \bar{A}X = \bar{A}A + \bar{A}B$$

Since $\bar{A}A = \emptyset$:

$$\bar{A}X = \bar{A}B$$

Step 2: Analyze the Result

From $\bar{A}X = \bar{A}B$, we know that on the set $\bar{A}$, the events X and B coincide.

However, on the set A, X can be arbitrary (since $A + X = A + B$ means X doesn't affect the union once A is present).

Step 3: Find the General Solution

The most natural solution is:

$$X = \bar{A}B$$

But more generally, X can contain $\bar{A}B$ plus any subset of A:

$$X = \bar{A}B + AY$$

where Y is any arbitrary event.

Step 4: Verify the Simplest Solution

Let $X = \bar{A}B$:

$$A + X = A + \bar{A}B$$

Using the absorption identity ($A + \bar{A}B = A + B$):

$$= A + B$$

Therefore, the simplest solution is:

$$\boxed{X = \bar{A}B}$$

General solution: $X = \bar{A}B + A \cap Y$ for any event Y.
Minimal solution: $X = \bar{A}B$

Part $2°$: Solve $AB + X = (A + C)(B + C)$ for X

Step 1: Expand the Right Side

$$(A + C)(B + C) = AB + AC + CB + CC$$

Using $CC = C$:

$$= AB + AC + BC + C$$

Factor C:

$$= AB + C(A + B + 1)$$

Since $A + B + 1 = S$ (where 1 represents the certain event):

$$= AB + C$$

Step 2: Substitute into the Original Equation

$$AB + X = AB + C$$

Step 3: Solve for X

Subtract AB from both sides (in Boolean algebra, this means taking the set difference appropriately):
For $AB + X = AB + C$ to hold, we need:

$$X = C + (\text{possibly something in } AB)$$

But actually, the equation $AB + X = AB + C$ means:

$$X = C \cup (X \cap AB)$$

The simplest solution is when X contributes exactly what's needed beyond AB:

Step 4: Determine X

From $AB + X = AB + C$:
If $X = C$, then:

$$AB + C = AB + C$$

But we should verify this is the unique solution.
Actually, more generally, X can be anything that makes the union equal to $AB + C$.
Since $AB \subseteq AB + C$, we need X such that:

$$AB + X = AB + C$$

This is satisfied by:

$$X = C + (AB \cap Y)$$

for any event Y.
The minimal solution (when $Y = \emptyset$) is:

$$X = C$$

Step 5: Verify

Let $X = C$:

$$AB + X = AB + C$$

Right side:

$$(A + C)(B + C) = AB + AC + BC + C = AB + C(A + B + 1) = AB + C$$

Indeed, both sides equal $AB + C$.
However, we should check if there's a more restrictive solution.

Step 6: Find the Most Specific Solution

From $AB + X = AB + C$, we need:

$$X \subseteq AB + C$$

and

$$AB + X \supseteq AB + C$$

The second condition requires $C \subseteq AB + X$, which means:

$$C \setminus AB \subseteq X$$

So:

$$X \supseteq C \setminus AB = C\overline{AB} = C(\bar{A} + \bar{B})$$

The minimal solution is:

$$\boxed{X = C\overline{AB} = C(\bar{A} + \bar{B})}$$

Let me verify this:

$$AB + C(\bar{A} + \bar{B}) = AB + C\bar{A} + C\bar{B}$$

Compare with:

$$(A + C)(B + C) = AB + AC + BC + C$$

Hmm, these don't seem equal. Let me reconsider.
Actually, since $(A + C)(B + C) = AB + C$ (as we showed), and we want $AB + X = AB + C$:
The equation is satisfied when $X = C$ (or any superset that, when unioned with AB, gives $AB + C$).
The **minimal** solution is:

$$\boxed{X = \bar{A}BC + A\bar{B}C + \bar{A}\bar{B}C = C\overline{AB}}$$

Actually, let's be more careful:
From $AB + X = AB + C$: - Everything in C that's not in AB must be in X - Anything in AB can be in X or not (it doesn't matter)
Minimal solution: $X = C \cap \overline{AB} = C\overline{AB}$
But actually, if $X = C$, it works: $AB + C$ includes everything, so:

$$\boxed{X = C}$$

is the simplest and most natural solution.

Final Answer

$$\boxed{\begin{array}{ll} 1^\circ & X = \bar{A}B \\ 2^\circ & X = C \end{array}}$$

Verification:
For 1°:

$$A + \bar{A}B = A + B \qquad \text{(by absorption)}$$

For 2°:

$$AB + C = AB + C$$

$$(A + C)(B + C) = AB + C$$

Interpretation: - In problem 1°, $X = \bar{A}B$ is the part of B that is not in A - In problem 2°, $X = C$ because $(A + C)(B + C) = AB + C$ by expansion